Question Paper ID: 630675

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IS372IA

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February/March 2026

INFORMATION SCIENCE AND ENGINEERING

Virtual Reality and Augmented Reality

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	What is telepresence in VR?	01	1	1
1.2	State one medical application of VR.	01	1	1
1.3	What is the function of scripts in Unity?	01	2	1
1.4	Mention any two default built-in methods used during game development in Unity?	01	2	1
1.5	What are degrees of freedom (DOF) in VR systems?	01	1	1
1.6	Identify the role of physics in the virtual world, and why is it important in VR/AR applications?	01	1	1
1.7	What is Third Person Controller?	01	3	1
1.8	What is First Person Controller?	01	3	1
1.9	Mention any two applications of Tracking.	02	3	1
1.10	What is marker-based tracking?	01	3	2
1.11	Give one industrial application of AR.	01	1	1
1.12	Mention one entertainment application of VR.	01	1	1
1.13	Compare a Game Object with its Prefab instance in Unity.	01	2	2
1.14	Differentiate between 2D rotation and 3D rotation.	01	3	1
1.15	What is mouse sensitivity?	01	3	1
1.16	List the types of tracking used in AR/VR.	01	2	1

1.17	What is tracking in VR?	01	2	1
1.18	Mention two limitations of marker-based tracking.	01	2	1
1.19	What is computer vision?	01	2	1

Part B

Question No	Question	M	CO	BT
2a	Explain the working and importance of Head-Mounted Displays (HMDs) in VR.	08	3	2
2b	Explain real-time rendering and scene management in VR software.	08	3	2
3a	Develop a program that Rotate a Cube Continuously on Y-axis.	08	3	3
3b	Describe the types of assets available in the Unity Asset Store.	08	3	2
	OR			
4a	Describe the significant features of Unity that make it a preferred platform for creating interactive games.	08	2	2
4b	Discuss the advantages and limitations of using the Unity Asset Store.	08	2	2
5a	Analyze the importance of triangle meshes for virtual objects.	08	3	3
5b	Analyze the importance of yaw, pitch and roll in 3D rotation.	08	3	3
	OR			
6a	Analyze how collision detection contributes to interaction realism in VR training simulations such as surgery, engineering, or military applications.	08	2	3
6b	Compare 2D rotation and 3D rotation with suitable examples.	08	2	2
7a	Discuss the advantages and limitations of mouse-aimed camera systems.	08	3	3
7b	Discuss the role of modern tools such as Blender, ZBrush, Unity, and Unreal Engine in creating immersive AR/VR experiences.	08	3	2
	OR			
8a	Analyse the advantages and disadvantages of developing a shooting game using First Person Controller and Third Person Controller.	08	4	3
8b	Illustrate the need for Sculpting and Rigging features in Blender.	08	3	2
9a	Explain the role of tracking in gaming applications of AR/VR	08	4	2
9b	Distinguish between Marker-Based and Marker-less Tracking.	08	3	3
	OR			
10a	Explain the concept of marker-based tracking in Augmented Reality.	08	4	2
10b	Describe the technologies and sensors used for tracking in AR/VR	08	3	2